Quantitative Data Analysis

Session 2 — Statistical Tests: Chi-square & Correlation

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Course code 2627_ECO_2_EN_009 · 2026–2027

Where we are

Session	Method	Question it answers
1	Univariate descriptive analysis	What does my data look like?
2	**Statistical tests: chi-square &**	**Are two variables related?**
today	**Pearson correlation**	
3	ANOVA + post-hoc tests	Do groups differ from each other?
4	Simple & multiple regression	What drives my outcome?

Last week ended on a limit: **univariate analysis cannot establish a relationship.**
Today we cross that line. From now on, every result you report is the answer to a **test**.

Today: 3 hours 30

Time	What happens	Mode
0:00 – 0:15	Dataset check: is your `.omv` clean and correctly typed?	Checkpoint
0:15 – 1:00	**Theory:** the logic of a statistical test & the map of all tests	Lecture + quiz
1:00 – 1:30	**Chi-square** test of independence + Cramér's V	Lecture
1:30 – 1:45	*Break*	
1:45 – 2:15	**Pearson & Spearman** correlation	Lecture
2:15 – 3:00	**Jamovi lab:** run both tests on your own dataset	Teams of 3
3:00 – 3:30	Write-up in protocol format + submission of Artefact 2	Checkpoint

You leave today with two tested relationships from your own data: one chi-square, one correlation, each written up as a managerial conclusion.

The logic of a statistical test

Why we need a test at all

In our retail sample, **71%** of customers aged 55+ buy the regional brand, against **42%** of those aged 18–34.

Is that a **real difference in the population** of shoppers… or just the luck of **these 325 people**?

A different sample would give slightly different figures. This is **sampling fluctuation**.

A statistical test answers one question:

Is the pattern I see too large to be explained by chance alone?

Yes → we generalise to the population.
No → we say nothing beyond our sample.

Descriptive statistics describe *your sample*. Tests let you speak about **the market**.

Two hypotheses, stated in business language

H_0 — the null hypothesis
Nothing is going on.
No relationship, no difference, no effect.
It is the hypothesis the test **tries to reject**.

H_1 — the alternative hypothesis
Something is going on.
There is a relationship / a difference.
Usually what the manager wants to know.

Managerial question	H_0	H_1
Does brand choice depend on age?	Brand and age band are **independent**	Brand and age band are **related**
Do frequent visitors spend more?	**No linear correlation** between visits and spending	**A correlation exists**

Exam habit: never write "H_0: $\rho = 0$" alone. Always translate into the manager's words.

The p-value and the 5% rule

p-value = the probability of observing a pattern at least this strong **if H_0 were true**.

α **= 5%** = the risk we accept of rejecting H_0 wrongly. Fixed *before* looking at the result.

In Jamovi the p-value is in the column **p**.
"$< .001$" means $p < 0.001$ — never write $p = 0$.

$p < 0.05$
Reject $H_0 \rightarrow$ the relationship is **statistically significant**.

$p \geq 0.05$
We **cannot reject** $H_0 \rightarrow$ no evidence of a relationship.

Remember Shapiro–Wilk (Session 1): there H_0 = "the data are normal", so $p > 0.05$ was the *good* news. For every test today, $p < 0.05$ is the finding.

Two ways to be wrong

	In reality: H_0 true (no relationship)	**In reality: H_0 false** (a relationship exists)
We reject H_0	**Type I error (α)** False alarm: we launch a targeted campaign for a segment effect that does not exist	**Correct**
We do not reject H_0	**Correct**	**Type II error (β)** Missed opportunity: a real segment effect goes unnoticed

We control the first risk by fixing $\alpha = 5\%$. The second one shrinks as the **sample size grows** — one more reason for your $N \geq 100$ rule.

Significance is not strength

The p-value answers:
Is there a relationship?
Yes / no, with a 5% risk.

The effect size answers:
Does it matter for the business?
Weak, moderate or strong.

Test	Effect size	Range	Session
Chi-square	Cramér's V	0 to 1	2
Pearson correlation	r (and r^2)	-1 to $+1$	2
ANOVA	η^2 (eta squared)	0 to 1	3
Regression	R^2, coefficients B	0 to 1	4

With $N = 5\,000$, a trivial correlation of $r = 0.04$ is "significant". **Always report both numbers.**

The protocol — same template, now with a test inside

1. **IDENTIFY** — Which test? Justify from the **nature of the variables**. Check its conditions.
2. **HYPOTHESISE** — State H_0 and H_1 in business language.
3. **DECIDE** — Report the statistic, df and p. Is $p < 0.05$? Then: **how strong** is the effect, and in which direction?
4. **RECOMMEND** — What does the manager do on Monday morning?

Step 1 now has a second half: **every test has conditions**. A correct test on data that violate its conditions is a wrong answer.

The map of statistical tests

Five questions that choose the test for you

1. **What is my objective?** Relate two variables, compare groups, or explain an outcome?
2. **How many variables?** One, two, or several at once?
3. **What type is each variable?** Nominal, ordinal, discrete, continuous — **Session 1 taxonomy**.
4. **How many groups, and are they independent?** Two groups or three and more? Different people, or the same people measured twice?
5. **Are the conditions met?** Normality, sample size, expected counts → **parametric** or **non-parametric** test.

Answer the five questions and the test is **mechanically determined**. There is no taste involved.

Parametric and non-parametric tests

Parametric

- Quantitative outcome (interval/ratio)
- Assumes an approximately **normal** distribution — or a large sample
- Works on **means** and variances
- More **powerful** when conditions hold

t-tests, ANOVA, Pearson, regression

Non-parametric

- Nominal or ordinal data, or skewed quantities
- **No normality** assumption
- Works on **counts** or **ranks**
- Robust to outliers; slightly less powerful

chi-square, Spearman, Mann–Whitney, Kruskal–Wallis

Your Session 1 normality check (three centres, skewness/kurtosis, histogram, Shapiro–Wilk) is exactly what decides between the two columns.

The full map — all the tests you should recognise

Situation	Parametric test	Non-parametric	Jamovi menu
Two qualitative variables	—	**Chi-square** + Cramér's V	Frequencies → Independent Samples
Two quantitative variables	**Pearson correlation (r)**	Spearman (ρ)	Regression → Correlation Matrix
Quantitative × 2 independent groups	Independent-samples t-test	Mann–Whitney U	T-Tests → Independent Samples
Quantitative × 3+ groups	One-way ANOVA + post-hoc	Kruskal–Wallis	ANOVA → One-Way ANOVA
Same people measured twice	Paired-samples t-test	Wilcoxon signed-rank	T-Tests → Paired Samples
One mean vs a target value	One-sample t-test	Wilcoxon (one sample)	T-Tests → One Sample
One qualitative vs expected split	—	χ^2 goodness of fit	Frequencies → N Outcomes
Several predictors → quantitative Y	Multiple regression	—	Regression → Linear Regression

Highlighted = today. ANOVA (session 3) and regression (session 4) build on the same logic.

Quiz — which test?

1. Is the choice of loyalty card (none / silver / gold) related to the customer's region?
2. Do firms with higher turnover also emit more CO_2?
3. Do online and in-store customers have the same average basket?
4. Is ESG grade (A to D) associated with the credit rating (AAA to B)?
5. Does satisfaction (0–20) differ across 4 store formats?
6. The same 40 employees rate their motivation before and after training.
7. Is our customer base split 50/50 between men and women, as in the city?
8. Which of ROI, headcount and debt ratio drive net profit?

Take 4 minutes. Use the five questions, then the map.

Quiz — answers

#	Test	Why / trap
1	Chi-square + Cramér's *V*	Nominal $\times$ nominal. Card level is ordinal, but χ^2 still applies.
2	Pearson (or Spearman)	Two continuous variables. Both are right-skewed: **check, then consider Spearman**.
3	Independent-samples t-test	Quantitative outcome, 2 independent groups.
4	Spearman correlation	Two **ordinal** variables: ranks, not means. χ^2 is acceptable but ignores the order.
5	One-way ANOVA + post-hoc	Quantitative outcome, 4 groups — **Session 3**.
6	Paired-samples t-test	The **same** people twice: the samples are not independent.
7	χ^2 goodness of fit	One qualitative variable against an expected distribution.
8	Multiple regression	Several predictors at once — **Session 4**.

Questions 2 and 4 are the ones students miss: **the conditions and the measurement level both matter**.

Two qualitative variables: chi-square

The question

Retail case (same data as Session 1, $N = 325$). The regional brand "Grain d'Or" is bought by 56% of customers. Marketing wants to know whether to **target an age group**.

Variable 1: age band — 18–34 / 35–54 / 55+
qualitative, 3 categories

Variable 2: brand purchased — National / Grain d'Or
qualitative (nominal), 2 categories

Qualitative $\times$ qualitative
$\rightarrow$ Chi-square test of independence
$\rightarrow$ effect size: Cramér's V

Question: is brand choice related to age?

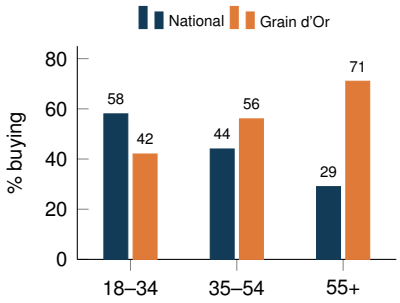
That sentence is your **IDENTIFY** step. It costs one line and earns a point.

Start with the contingency table

Observed counts (row %)

Age band	National	Grain d'Or	Total
18–34	58 (58%)	42 (42%)	100
35–54	55 (44%)	70 (56%)	125
55+	29 (29%)	71 (71%)	100
Total	142	183	325

Read **row percentages**: they compare age bands of different sizes fairly.



The pattern is visible: the older the customer, the more regional brand. **The test tells us whether it is more than chance.**

What the chi-square actually does

1. It builds the table you would expect **if H_0 were true** — if brand did not depend on age:

$$E = \frac{\text{row total} \times \text{column total}}{N}$$

e.g. 18–34 & Grain d'Or: $100 \times 183/325 = 56.3$ expected, but only **42** observed.

2. It adds up the gaps between observed and expected:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Expected counts under H_0

Age band	National	Grain d'Or
18–34	43.7	56.3
35–54	54.6	70.4
55+	43.7	56.3

Big gaps → big χ^2 → small p.
Here the 35–54 row matches almost perfectly; the **gaps are at both ends**.

You will never compute this by hand. You need it to explain what the number means.

Conditions of the chi-square test

1. **Counts, not percentages.** The test works on frequencies of individuals.
2. **Independent observations.** Each person appears in one cell only.
3. **Expected counts large enough:** at least **80% of cells with** $E \geq 5$, and none below 1.

Our case: smallest expected count = 43.7. Condition met.

If violated: merge small categories (e.g. "AAA" + "AA") or, for a 2×2 table, use **Fisher's exact test**.

Your nominal variable must have **2–5 categories** — this is why that rule was in the dataset specification.

Hypotheses, degrees of freedom, decision

H_0**:** brand choice is independent of age band.
H_1**:** brand choice depends on age band.

Degrees of freedom

$$df = (\text{rows} - 1)(\text{columns} - 1) = (3-1)(2-1) = 2$$

A 2×2 table **always** has $df = 1$. Remember this — we will use it in five minutes.

χ^2 **Tests** (Jamovi)

	Value	df	p
χ^2	17.10	2	$< .001$
N	325		

$p < 0.001 < 0.05 \rightarrow$ reject H_0.
Brand choice is **significantly related** to age.

How strong? Cramér's V

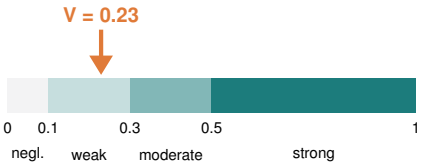
$$V = \sqrt{\frac{\chi^2}{N \times (k-1)}}$$

k = the smaller number of rows or columns.

Our case: $\sqrt{17.10/(325 \times 1)} = \mathbf{0.23}$

Nominal (Jamovi)

	Value
Phi-coefficient	0.229
Cramér's V	0.229



A **real but weak-to-moderate** association: age matters, but it is far from the whole story.

Direction? V has none. Go back to the **row percentages** to say who buys what.

Writing it up — model answer

1. IDENTIFY. Age band and brand purchased are both qualitative, so we use a chi-square test of independence, with Cramér's V for strength. All expected counts exceed 5 (minimum 43.7).

2. HYPOTHESISE. H_0: brand choice does not depend on age. H_1: brand choice depends on age.

3. DECIDE. $\chi^2(2) = 17.10$, $p < .001$: we reject H_0 at the 5% level. Cramér's $V = 0.23$ indicates a weak-to-moderate association. The share buying Grain d'Or rises from 42% (18–34) to 71% (55+).

4. RECOMMEND. Keep the core Grain d'Or message for customers aged 55+, who are already loyal to it; test a dedicated offer (digital coupons, sampling) to convert 18–34 year-olds, where the national brand still dominates.

Step 4 must follow *from the numbers*. "Improve marketing" earns nothing. "Target the segment where the gap is" does.

Read output critically — a real example

Test of association: gender $\times$ brand purchased (National / Regional), $N = 325$

	Value	df	p
χ^2	221.43	2	0.000
Cramér's V	0.765		
N	325		

Two things are wrong here.
Can you spot them before I tell you?

df = 2 — gender (2) $\times$ brand (2) is a 2×2 table: $df = 1$.
$V = 0.765$ — with $\chi^2 = 221.43$ and $N = 325$, $V = \sqrt{221.43/325} = 0.83$. The two numbers cannot come from the same table.

Same lesson as Session 1: software never refuses to print a number. Checking is your job.

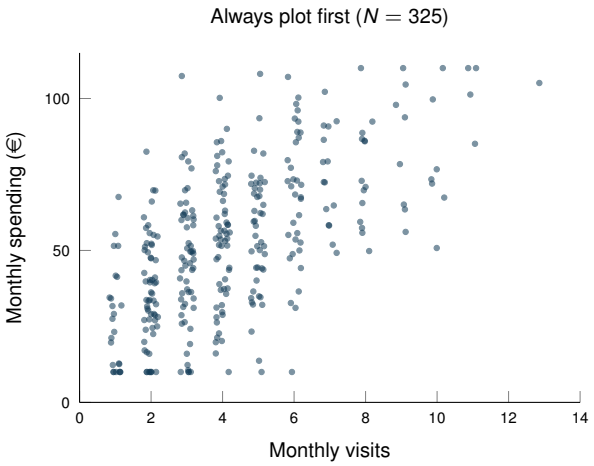
Two quantitative variables: correlation

The question

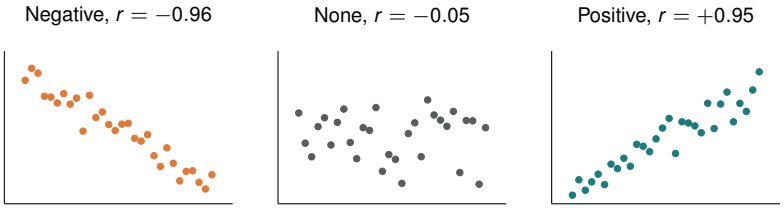
Same retail data. Does a customer who **visits more often** also **spend more** per month?

Variable 1: monthly visits (discrete)
Variable 2: monthly spending € (continuous, ratio)

Quantitative × quantitative
→ Pearson correlation r
(Spearman ρ if conditions fail)



Pearson's r: direction and strength



Sign = direction. **Size** = strength.
r runs from -1 (perfect negative) to $+1$ (perfect positive); 0 = no *linear* link.

$\lvert r \rvert$	Strength
< 0.3	weak
$0.3 - 0.7$	moderate
> 0.7	strong

The Jamovi output — and how to read it

Correlation Matrix

		Visits	Spending	Age
Visits	Pearson's r	—		
	p-value	—		
Spending	Pearson's r	**0.618**	—	
	p-value	**< .001**	—	
Age	Pearson's r	−0.099	−0.046	—
	p-value	0.075	0.413	—

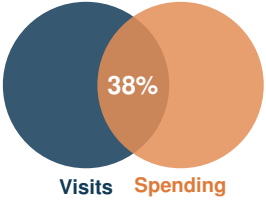
$df = N - 2 = 323$ for every pair.

H_0**:** no linear relationship between visits and spending.

Visits–spending: $r = 0.62$, $p < .001 \rightarrow$ reject H_0: a **moderate positive** correlation.

Age–spending: $r = -0.05$, $p = .413 \rightarrow$ cannot reject H_0: **no evidence** that age is linked to spending.

r^2: how much the two variables share



$$r^2 = 0.618^2 = 0.38$$

38% of the variation in monthly spending goes together with the variation in visits.

The other 62% depends on something else: basket size, income, promotions…

> This is the seed of R^2 in **Session 4**: regression will tell us *how many euros* one more visit is worth.

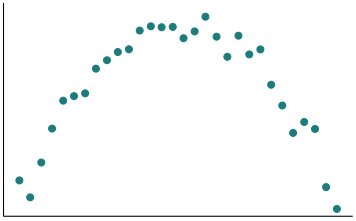
Conditions of Pearson's *r*

1. **Interval or ratio variables** — not ordinal codes, never nominal ones.
2. **A linear relationship** — check the scatterplot.
3. **Approximate normality** of both variables — your Session 1 evidence.
4. **No extreme outliers** — one point can create or destroy r.
5. **Homoscedasticity** — the spread of Y is similar along X (no "funnel").

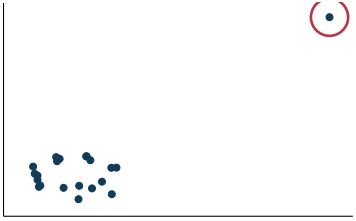
Our case: visits are right-skewed (skewness 0.90, Shapiro–Wilk $p < .001$). With $N = 325$, Pearson is robust — but we **confirm with Spearman**: $\rho = 0.61$, $p < .001$. Same conclusion, so the result is solid.

Two scatterplots that fool the coefficient

Strong curve, yet $r = 0.01$



One outlier: $r = 0.83$ (without it: -0.01)



Pearson only sees **straight lines** and is **hijacked by extreme points**. A coefficient without a scatterplot is not an analysis.

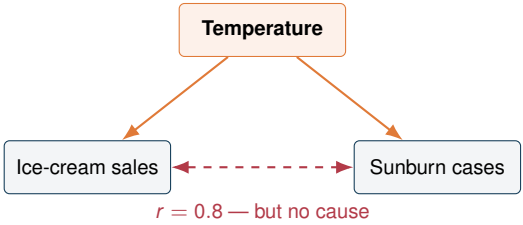
Pearson or Spearman?

	Pearson r	**Spearman** ρ
Variables	Interval / ratio	Ordinal, or skewed quantities
Relationship detected	Linear	Monotonic (always up or always down)
Normality	Needed (or large N)	Not needed
Outliers	Very sensitive	Robust — works on ranks
Reading	-1 to $+1$	-1 to $+1$, same rules

Use Spearman for two ordinal variables (ESG grade $\times$ credit rating) or for heavily skewed data (turnover, emissions).

Good practice: tick both in Jamovi. If they agree, report Pearson; if they disagree, trust Spearman and say why.

Correlation is not causation



A correlation can come from:

- a **third variable** driving both
- **reverse causality** (do visits cause spending, or do big spenders visit more?)
- pure **coincidence**

In your write-up say “is associated with”, never “causes” or “increases”. Isolating the effect of one variable, the others held constant, is the job of **regression in Session 4**.

Writing it up — model answer

1. IDENTIFY. Monthly visits and monthly spending are both quantitative, so we use Pearson's correlation. The scatterplot shows a linear pattern without extreme outliers; visits are right-skewed, so we check with Spearman.

2. HYPOTHESISE. H_0: there is no linear relationship between visits and spending. H_1: there is one.

3. DECIDE. $r(323) = 0.62$, $p < .001$: we reject H_0. The correlation is positive and moderate; Spearman confirms it ($\rho = 0.61$). Visits share 38% of their variance with spending. Age is not related to spending ($r = -0.05$, $p = .41$).

4. RECOMMEND. Visit frequency, not age, is the lever: a loyalty mechanism that rewards the next visit (points expiring monthly, weekly offers) is a better bet than age-targeted promotions. Session 4 will quantify the euro value of one extra visit.

Jamovi lab

Chi-square in Jamovi

Analyses → Frequencies → Independent Samples (χ^2 test of association)

1. Move one variable to **Rows**, the other to **Columns**
2. **Statistics:** tick χ^2 and **Phi and Cramér's V**
3. **Cells:** tick Observed and **Expected** counts, and **Row** percentages
4. Optional: **Plots** → bar plot, percentages within rows

Read, in this order:
1. Expected counts — condition OK?
2. χ^2 Tests — value, df, p
3. Nominal — Cramér's V
4. Row % — who does what?

Both variables must be set to **Nominal** or **Ordinal**. A continuous variable dropped here gives a table with one column per value — and a meaningless test.

Correlation in Jamovi

Analyses → Regression → Correlation Matrix

1. Move **all your continuous variables** in at once (at least 3)
2. Correlation coefficients: tick **Pearson** and **Spearman**
3. Additional options: **Report significance**, **Flag significant correlations**
4. Plot: tick **Correlation matrix** to see the scatterplots

For one pair in detail: **Exploration → Scatterplot**, add a linear regression line.

Read, in this order:
1. The scatterplot — linear? outliers?
2. p — significant?
3. r — sign and strength
4. Spearman — does it agree?

Firm-level data? Turnover and emissions are usually very skewed. Rely on **Spearman**, or create a log variable via **Compute**.

Apply it to your fil rouge dataset

Chi-square pair
Two qualitative variables, 2–5 categories each.
Firm data: sector × ESG grade band
Retail data: region × brand purchased

Correlation set
At least three continuous variables.
Firm data: turnover, ROI, emissions per employee
Retail data: visits, spending, basket size

For each pair, before you touch Jamovi, write one line: **“A manager would care because…”**
If you cannot finish that sentence, pick another pair.

A non-significant result is **still a result**. “Sector is not related to ESG grade” is useful news — report it honestly.

Artefact 2 — submit before you leave (25 marks)

1. The updated `.omv` file, with any new cleaning added to your **cleaning log**
2. **One chi-square**: contingency table with row %, check of expected counts, χ^2, df, p, Cramér's V
3. **One correlation matrix** of at least three continuous variables, with the scatterplot of the pair you discuss, Pearson and Spearman
4. Each test written up in the **four-step protocol**
5. **Two managerial recommendations**, each tied to a specific number

Indicative weighting: chi-square 10 · correlation 10 · recommendations 5.
As in Artefact 1, **half the marks are for interpretation**. Output alone is not analysis.

Errors that cost marks every year

- Running a **correlation on nominal codes** — "r between sector and region = 0.12"
- Reporting χ^2 with **percentages instead of counts**
- Ignoring the **expected-count** condition when categories are small
- Reading a significant p as a **strong** effect — report V or r every time
- Writing "visits **cause** higher spending" — a correlation shows association only
- Reporting r **without looking at the scatterplot**
- Stating "H_0 is proven" — we only **fail to reject** it

Before session 3

1. **Finish Artefact 2** if your write-up is not complete.
2. Identify in your dataset a **qualitative variable with 3–5 groups** and a **quantitative outcome**.
   e.g. sector → ROI; store format → satisfaction; region → spending
3. Write the managerial question: **do the groups differ on this outcome?**
4. Check the outcome's distribution **within each group** (Descriptives → split by).
5. Read: Navarro & Foxcroft, *Learning Statistics with Jamovi*, chapters on categorical data analysis and correlation.

Today you asked *are two variables related?* Next week: **do groups differ — and which ones?**

Questions?

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Session 3: comparing groups — ANOVA & post-hoc tests